

1. Elevators have many states, in the simplest terms, including running state, stopping state, opening state and closing state. And four operations can be performed: operation “stop”, “open”, “run” and “close”. State pattern is useful with this scenario. Please write the code to implement the pattern.

Client

```
package experiment20_1;

public class Client {

    public static void main(String[] args) {

        Lift list= new Lift ();

        list.open();

        list.close();

        list.run();

        list.stop();

    }

}
```

Close

```
package experiment20_1;

class Close implements LiftState{

    @Override

    public void running(Lift lift) {

        System.out.println("The elevator is running.");

        lift.setLiftState(new Run());

    }

}
```

```
}

@Override
public void stopping(Lift lift) {
    System.out.println("The elevator is stopping.");
    lift.setLiftState(new Stop());
}

@Override
public void opening(Lift lift) {
    System.out.println("The elevator door is opening.");
    lift.setLiftState(new Open());
}

@Override
public void closing(Lift lift) {
    System.out.println("The elevator door is already
closing.");
}
}
```

Lift

```
package experiment20_1;

class Lift {

    private LiftState liftState;

    public Lift(){

        this.liftState = new Close();

    }

    public void setLiftState(LiftState liftState){

        this.liftState = liftState;

    }

    public void run(){

        liftState.running(this);

    }

    public void stop(){

        liftState.stopping(this);

    }

    public void open(){

        liftState.opening(this);

    }

    public void close(){

        liftState.closing(this);

    }

}
```

```
}  
}
```

LiftState

```
package experiment20_1;  
  
interface LiftState {  
    void running(Lift lift);  
    void stopping(Lift lift);  
    void opening(Lift lift);  
    void closing(Lift lift);  
}
```

Open

```
package experiment20_1;  
  
class Open implements LiftState{  
    @Override  
    public void running(Lift lift) {  
        System.out.println("The elevator is running.");  
        lift.setLiftState(new Run());  
    }  
  
    @Override
```

```

    public void stopping(Lift lift) {
        System.out.println("The elevator is stopping.");
        lift.setLiftState(new Stop());
    }

    @Override
    public void opening(Lift lift) {
        System.out.println("The elevator door is already
opening.");
    }

    @Override
    public void closing(Lift lift) {
        System.out.println("The elevator door is closing.");
        lift.setLiftState(new Close());
    }
}

```

Run

```

package experiment20_1;

class Run implements LiftState{
    @Override

```

```
public void running(Lift lift) {  
    System.out.println("The elevator is already running.");  
}
```

@Override

```
public void stopping(Lift lift) {  
    System.out.println("The elevator is stopping.");  
    lift.setLiftState(new Stop());  
}
```

@Override

```
public void opening(Lift lift) {  
    System.out.println("The elevator door is opening.");  
    lift.setLiftState(new Open());  
}
```

@Override

```
public void closing(Lift lift) {  
    System.out.println("The elevator door is closing.");  
    lift.setLiftState(new Close());  
}
```

```
}
```

Stop

```
package experiment20_1;

class Stop implements LiftState{

    @Override

    public void running(Lift lift) {

        System.out.println("The elevator is running.");

        lift.setLiftState(new Run());

    }


    @Override

    public void stopping(Lift lift) {

        System.out.println("The elevator is already stopped.");

    }


    @Override

    public void opening(Lift lift) {

        System.out.println("The elevator door is opening.");

        lift.setLiftState(new Open());

    }


    @Override
```

```
public void closing(Lift lift) {  
    System.out.println("The elevator door is closing.");  
    lift.setLiftState(new Close());  
}  
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ\bin\idea_rt.jar" D:\JDK\bin\java.exe  
The elevator door is opening.  
The elevator door is closing.  
The elevator is running.  
The elevator is stopping.  
  
Process finished with exit code 0
```